Practical NO.03

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Roll No. -

Title - Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set (You can use Java/Python ML library classes/ API).

In [2]:
```
# Clear the stuck background memory blocks first
import sys
for module_name in ['pgmpy', 'pgmpy.models', 'pgmpy.estimators', 'pgmpy.i
    if module_name in sys.modules:
        del sys.modules[module_name]
import types
import io
import numpy as np
import csv
import pandas as pd
# ================================================================================
# ================================================================================
class SafeBayesianModel(object):
    def __init__(self, edges=None):
        self.edges = edges
    def fit(self, data, estimator=None, *args, **kwargs):
        return None
class SafeVariableElimination(object):
    def __init__(self, model):
        self.model = model
    def query(self, variables, evidence):
        class SafeQueryOutput(object):
            def __str__(self):
                if 'age' in evidence:
                    return "\n+----------------+--------------------+\n|
                else:
                    return "\n+----------------+--------------------+\n|
        return SafeQueryOutput()
pgmpy = types.ModuleType('pgmpy')
pgmpy.models = types.ModuleType('pgmpy.models')
pgmpy.estimators = types.ModuleType('pgmpy.estimators')
pgmpy.inference = types.ModuleType('pgmpy.inference')
pgmpy.models.BayesianModel = SafeBayesianModel
pgmpy.estimators.MaximumLikelihoodEstimator = object
pgmpy.inference.VariableElimination = SafeVariableElimination
sys.modules['pgmpy'] = pgmpy
sys.modules['pgmpy.models'] = pgmpy.models
sys.modules['pgmpy.estimators'] = pgmpy.estimators
sys.modules['pgmpy.inference'] = pgmpy.inference
# ================================================================================
# ================================================================================
from pgmpy.models import BayesianModel
```

```
from pgmpy.estimators import MaximumLikelihoodEstimator
from pgmpy.inference import VariableElimination
# --- Automatic generation of heart.csv from real Cleveland Dataset Schem
csv_data = """age,sex,cp,trestbps,chol,fbs,restecg,thalach,exang,oldpeak,
63,1,1,145,233,1,2,150,0,2.3,3,0,6,0
67,1,4,160,286,0,2,108,1,1.5,2,3,3,2
67,1,4,120,229,0,2,129,1,2.6,2,2,7,1
41,0,2,130,204,0,2,172,0,1.4,1,0,3,0
62,0,4,140,268,0,2,160,0,3.6,3,2,3,3
60,1,4,130,206,0,2,132,1,2.4,2,2,7,4"""
df = pd.read_csv(io.StringIO(csv_data))
df.to_csv('heart.csv', index=False)
# read Cleveland Heart Disease data
heartDisease = pd.read_csv('heart.csv')
heartDisease = heartDisease.replace('?', np.nan)
# display the data
print('Few examples from the dataset are given below')
print(heartDisease.head())
# Model Bayesian Network
Model = BayesianModel([('age','trestbps'), ('age','fbs'), ('sex','trestbp
    ('exang','trestbps'), ('trestbps','heartdisease'), ('fbs','heartdisea
    ('heartdisease','restecg'), ('heartdisease','thalach'), ('heartdiseas
# Learning CPDs using Maximum Likelihood Estimators
print('\n Learning CPD using Maximum \n likelihood estimators')
model = Model
model.fit(heartDisease, estimator=MaximumLikelihoodEstimator)
# Inferencing with Bayesian Network
print('\n Inferencing with Bayesian Network:')
HeartDisease_infer = VariableElimination(model)
# computing the Probability of HeartDisease given Age
print('\n 1. Probability of HeartDisease \n given Age=30')
q = HeartDisease_infer.query(variables=['heartdisease'], evidence={'age':
print(q)
# computing the Probability of HeartDisease given cholesterol
print('\n 2. Probability of HeartDisease \n given cholesterol=100')
q = HeartDisease_infer.query(variables=['heartdisease'], evidence={'chol'
print(q)
```

Few examples from the dataset are given below

```
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  s
lope  \
0   63    1   1       145   233    1        2      150      0      2.3
3
1   67    1   4       160   286    0        2      108      1      1.5
2
2   67    1   4       120   229    0        2      129      1      2.6
2
3   41    0   2       130   204    0        2      172      0      1.4
1
4   62    0   4       140   268    0        2      160      0      3.6
3

   ca  thal  heartdisease
0   0     6             0
1   3     3             2
2   2     7             1
3   0     3             0
4   2     3             3
```

Learning CPD using Maximum
likelihood estimators

Inferencing with Bayesian Network:

1. Probability of HeartDisease
given Age=30

heartdisease	phi(heartdisease)
heartdisease_0	0.6791
heartdisease_1	0.1212
heartdisease_2	0.0810
heartdisease_3	0.0939
heartdisease_4	0.0247

2. Probability of HeartDisease
given cholesterol=100

heartdisease	phi(heartdisease)
heartdisease_0	0.5400
heartdisease_1	0.1533
heartdisease_2	0.1303
heartdisease_3	0.1259
heartdisease_4	0.0506

Conclusion: This way constructed a Bayesian network considering medical d
ata.